

 PlayStation.

S T Q Q S S D

1) Para 8 bits

$$\begin{array}{r} 1 \text{ L } 2 \\ 1 \text{ O} \\ 1 \rightarrow 0 + 1 \end{array}$$

Resposta = 00000001

118) para 8 bits

$$\begin{array}{r} 118 \text{ L } 2 \\ 0 \text{ 59 L } 2 \\ 1 \text{ 29 L } 2 \\ 1 \text{ 14 L } 2 \\ 0 \text{ 7 L } 2 \\ 1 \text{ 3 L } 2 \\ 1 \text{ 1 L } 2 \\ 1 \text{ 0} \end{array}$$

$$1110110 \rightarrow 0001001$$

Resposta = 00001010

2800) para 8 bits

$$\begin{array}{r} 2800 \text{ L } 2 \\ 0 \text{ 1400 L } 2 \\ 0 \text{ 700 L } 2 \\ 0 \text{ 350 L } 2 \\ 0 \text{ 175 L } 2 \\ 1 \text{ 87 L } 2 \\ 1 \text{ 43 L } 2 \\ 1 \text{ 21 L } 2 \\ 1 \text{ 10 L } 2 \\ 0 \text{ 5} \end{array}$$

5 L 2

$$1 \text{ 2 L } 2$$

$$0 \text{ 1 L } 2$$

$$1 \text{ 0}$$

spiral



21) demonstre os complementos de dois dos números.

a) 123 (para 8 bits)

123 / 2

1 61 / 2

1 30 / 2

0 15 / 2

1 7 / 2

1 3 / 2

1 1 / 2

1 0

1111011

0000100

+

Resposta = 0000101

b) 50 - 51 (8 bits)

50 / 2

0 25 / 2

1 12 / 2

0 6 / 2

0 3 / 2

1 1 / 2

1 0

Resposta = 110010

51 / 2

1 25 / 2

1 12 / 2

0 6 / 2

0 3 / 2

1 1 / 2

1 0

Resposta = 110011 → 001100

+

001101

110010

+ 001101

111111

Não ocorreu overflow

c) 120 - 5 (8 bits)

$$\begin{array}{r}
 120 \text{ L } 2 \\
 0 \ 60 \text{ L } 2 \\
 0 \ 30 \text{ L } 2 \\
 0 \ 15 \text{ L } 2 \\
 1 \ 7 \text{ L } 2 \\
 1 \ 3 \text{ L } 2 \\
 1 \ 1 \text{ L } 2 \\
 1 \ 0
 \end{array}$$

Resposta = 1111000

$$\begin{array}{r}
 5 \text{ L } 2 \\
 1 \ 2 \text{ L } 2 \\
 0 \ 1 \text{ L } 2 \\
 1 \ 0
 \end{array}$$

Resposta = 101 + 0000101

$$\begin{array}{r}
 1111000 \\
 + 1111011 \\
 \hline
 1000011
 \end{array}$$

Ocorreu Overflow

d) 150 - 50 (8 bits)

$$\begin{array}{r}
 150 \text{ L } 2 \\
 0 \ 25 \text{ L } 2 \\
 1 \ 12 \text{ L } 2 \\
 6 \ 3 \text{ L } 2 \\
 1 \ 1 \text{ L } 2 \\
 1 \ 0
 \end{array}$$

$$\begin{array}{r}
 150 = 110010 \\
 50 = 001110 \\
 \hline
 100000
 \end{array}$$

Ocorreu Overflow

Lista 3

1) Converta os números binários em decimal

a) $10011 = 1 + 2 + 0 + 0 + 32 = 35$

b) $111011,101 = 1 + 2 + 0 + 8 + 16 + 32 = 59 > 59,625$
 $0,5 + 0 + 0,125 = 0,625$

c) $111,11001 = 1 + 2 + 4 = 7$
 $0,5 + 0,25 + 0 + 0 + 0,03125 = 0,78125$

